

Subtitle Basic idea about DC to DC converter

Institute Dr B C Roy Engineering College, Durgapur

Subject Power Electronics PC-EE504

Calculation of Critical L and C of DC-DC Chopper

Kingsuk Majumdar

Hello friends

Lecture on Critical L & C Chopper

- ① Brief recap of chopper (say Buck Chopper)
- ② Calculation of critical L (for buck chopper)
- ③ Calculation of critical C (for buck chopper)
- ④ Signification of L_{cri} and C_{cri}
- ⑤ References

Brief recap of chopper (say Buck Chopper)

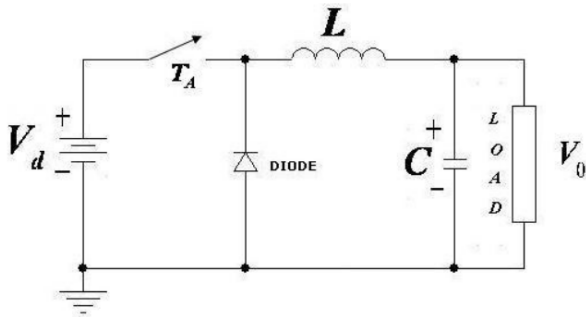


Figure: Buck chopper circuit ¹

¹picture is ©2013 K Majumdar

vltage and current wave of buck chopper

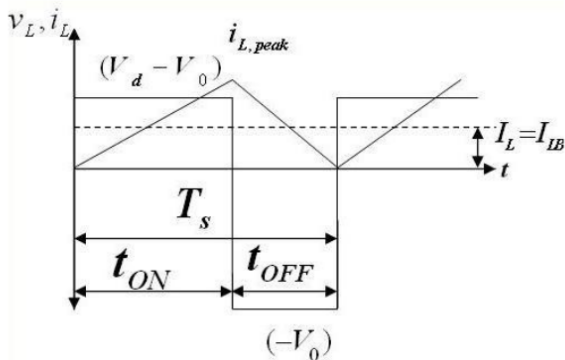


Figure: Inductor voltage and current in buck chopper in CCM in steady state condition

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²Continuous conduction mode

³picture is ©2013 K Majumdar

Calculation of critical inductance, L_{cri}

$$v_L = L \frac{di}{dt} \quad (1)$$

During t_{ON}

$$V_d - V_o = L \frac{\Delta I}{\Delta T} \quad (2)$$

$$\Delta I = \frac{V_d - V_o}{L} T_{ON} \quad (3)$$

During t_{OFF}

$$-V_o = L \frac{\Delta I}{\Delta T} \quad (4)$$

$$-V_o = L \frac{\Delta I}{T_{OFF}} \quad (5)$$

cont...

again in steady state operation in CCM

$$I_{LB} = \frac{1}{2} I_{L,peak} \text{ ,in steady state} \quad (6)$$

$$I_{o,ave} = \frac{1}{2} I_{L,peak} = I_o \quad (7)$$

and from Fig. 2,

$$\Delta I = \|I_2 - I_1\| = 2I_o \quad (8)$$

$$\frac{\Delta I}{I_o} = 2 \quad (9)$$

cont...

Now,

$$2I_o = \frac{1}{L_{cri}}(V_d - V_o)T_{ON} \quad (10)$$

$$L_{cri} = \frac{1}{2} \frac{1}{I_o} V_d (1 - D) D T_S \quad (11)$$

$$= \frac{1}{2} R \frac{V_d}{V_o} (1 - D) D T_S \quad (12)$$

$$L_{cri} = \frac{1}{2} R (1 - D) T_S, \text{ for buck chopper in CCM} \quad (13)$$

$$V_o = D V_d \text{ and } T_{ON} = D T_S \text{ and duty ratio i.e. } D = \frac{T_{ON}}{T_S}, \frac{V_o}{I_o} = R$$

Calculation of critical Capacitance, C_{cri}

In this section we will learn voltage ripple and calculation of critical capacitance.

$$V_{o,ave} = \frac{1}{2} \Delta V_c = V_o \quad (14)$$

$$Q = CV_o \quad (15)$$

$$\Delta Q = C \Delta V_o \quad (16)$$

Now,

$$\Delta Q = \frac{1}{2} \frac{\Delta I}{2} \frac{T_s}{2} \quad (17)$$

$$= \frac{1}{8} T_s (\Delta I) \quad (18)$$

cont...

Now from Eq. 11 and 18

$$\Delta Q = \frac{1}{8} T_s \left(\frac{1}{L_{cri}} V_o (1 - D) T_s \right) \quad (19)$$

$$\frac{\Delta V_o}{V_o} = \frac{1}{8} \frac{1}{CL_{cri}} (1 - D) T_s^2, \text{from Eq 16} \quad (20)$$

Again we know Cutoff frequency f_c is

$$f_c = \frac{1}{2\pi} \frac{1}{\sqrt{CL_{cri}}} \quad (21)$$

$$f_s = \frac{1}{T_s}, \text{ is the switching frequency} \quad (22)$$

cont...

From Eq. 20, 22 and 21

$$\frac{\Delta V_o}{V_o} = \frac{1}{2} \pi^2 (1 - D) \left(\frac{f_c}{f_s} \right)^2 \quad (23)$$

Hence, Eq. 23 is called voltage ripple equation for buck chopper. It is noticed that it is irrespective of load (R).

Now, from Eq. 20 we can write for critical capacitance,

$$C_{cri} = \frac{1}{8} \frac{1}{L_{cri}} \left(\frac{V_o}{\Delta V_o} \right) (1 - D) T_S^2 \quad (24)$$

$$= \frac{1}{8} \frac{1}{L_{cri}} \left(\frac{V_o}{2V_o} \right) (1 - D) T_S^2, \text{ as } \Delta V_o = 2V_o \quad (25)$$

$$= \frac{1}{16} \frac{1}{L_{cri}} (1 - D) T_S^2 \quad (26)$$

Signification of L_{cri} and C_{cri}

From Eq. 13, 23 and 26, we can conclude:

- choosing L_{cri} is depending on load. duty and switching frequency.
- for light load i.e. high value of R , L_{cri} will be high
- for higher switching frequency low L_{cri} will be needed
- chosen L i.e. L_{chs} must be in the range of $L_{cri} < L_{chs} < 3L_{cri}$
- f_s must be greater than f_c (at least 10 times) for better control and less voltage ripple.
- duty also makes a signification role on voltage ripple, $D < 0.4$ in buck in CCM
- switching frequency makes a huge impact on converter characteristics

Switching frequency impact on buck chopper

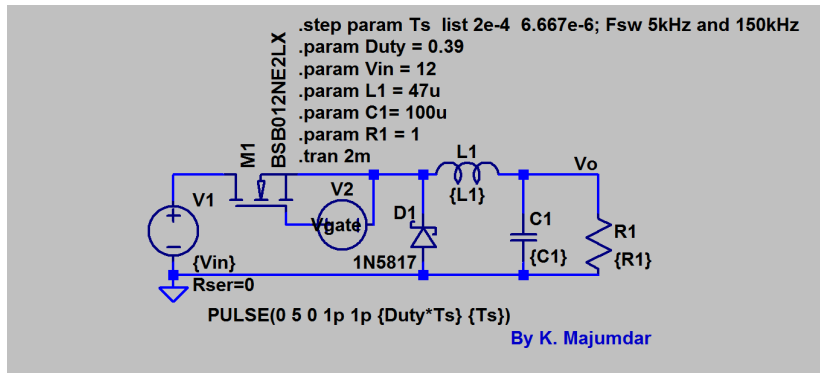


Figure: Buck chopper in CCM for 5kHz and 150kHz⁴

⁴<https://github.com/KingsukMajumdar/KMajumdar.git> , LTspice

response of output voltage at 5kHz and 150kHz

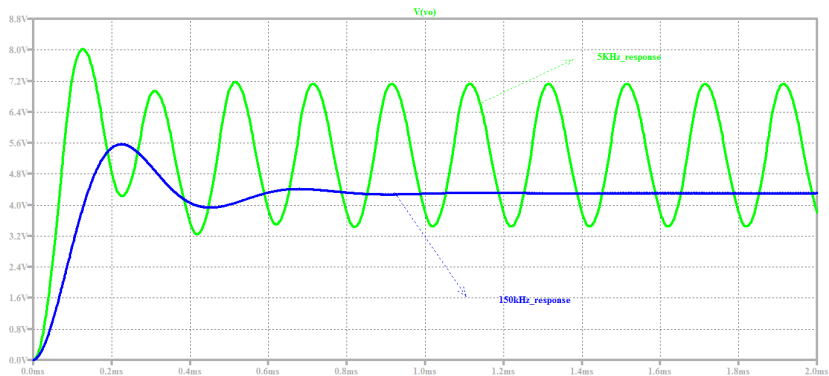


Figure: response of output voltage at 5kHz and 150kHz of buck chopper in CCM

References



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